Exam #1 Math 742 - Math Stats David Cortes

Part I

Problem 1. Let X have PDF $f(x) = \begin{cases} 2x : 0<x<1 \\ 0 : \text{Otherwise} \end{cases}$

(a) compute $\mathbb{E}[X]$ (a.1) and $V(X)$ (a.2)

(b) Let $Y = X^2$. Find the PDF of Y

(c) Verify Jensen's inequality for $\phi(x) = x^2$ by comparing $\mathbb{E}[X^2]$ and $(\mathbb{E}[X])^2$

(a)

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx = \int_0^1 x(2x)\,dx = 2\int_0^1 x^2\,dx = 2\left[\frac{x^3}{3}\right]_0^1 = 2\cdot\frac{1}{3} = \frac{2}{3} \rightarrow \mathbb{E}[X] = \frac{2}{3}$$ (a.1)

Now lets compute $\mathbb{E}[X^2] = \int_0^1 x^2(2x)\,dx = \int_0^1 2x^3\,dx = 2\left[\frac{x^4}{4}\right]_0^1 = 2\cdot\frac{1}{4} = \frac{1}{2}$

$$\text{Var}(x) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \rightarrow \text{Var}(x) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2}\,{}^{9}_{18} - \frac{4}{9}\,{}^{8}_{18} \rightarrow \text{Var}(x) = \frac{9}{18} - \frac{8}{18} = \frac{1}{18} \rightarrow \text{Var}(X) = \frac{1}{18}$$ (a.2)

(b)

Compute the CDF $F_Y(y) \rightarrow F_Y(y) = P(Y \le y) = P(X^2 \le y)$

$\rightarrow$ If $y \le 0$: since $Y>0$, $F_Y(y) = 0$

$\rightarrow$ If $0<y<1$: because $X>0$, $x^2 \le y \leftrightarrow x \le \sqrt{y}$

$F_Y(y) = P(X \le \sqrt{y}) = \int_0^{\sqrt{y}} 2x\,dx$

Compute $\int_0^{\sqrt{y}} 2x\,dx = \left[x^2\right]_0^{\sqrt{y}} = y$

for $0<y<1$, $F_Y(y) = y$

If $y \ge 1$: Since $Y<1$, $F_Y(y) = 1$

So the CDF is: $F_Y(y) = \begin{cases} 0, & y<0 \\ y, & 0<y<1 \\ 1, & y \ge 1 \end{cases}$

Now lets Differentiate to get the PDF (for $0<y<1$)

$$f_Y(y) = \frac{d}{\partial y} F_Y(y) = \frac{\partial}{\partial y}(y) = 1$$

Thus

$$f_Y(y) = \begin{cases} 1, & 0<y<1 \\ 0, & \text{otherwise} \end{cases}$$

$\rightarrow$ This is the uniform $(0,1)$ density (PDF $=1$ on $[0,1]$)

(c) Jensen (for convex ϕ) says $\phi(E[X]) \le E[\phi(X)]$.

Here $\phi(x) = x^2$, so it predicts $(E[X])^2 \le E[X^2]$

We computed $E[X] = \frac{2}{3} \rightarrow (E[X])^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$ and $E[X^2] = \frac{1}{2}$ compare $\frac{4}{9} \overset{?}{\le} \frac{1}{2}$

convert to decimals or common denominator: $\frac{4}{9} \approx 0.444$, $\frac{1}{2} = 0.5$, so indeed $\rightarrow$

$(\mathbb{E}[X])^2 = \frac{4}{9} \le \frac{1}{2} = E[X^2]$. $\rightarrow$ Using the identity $\text{Var}(x) = E[X^2] - (E[X])^2$ gives $E[X^2] - (E[X])^2 \ge 0 \rightarrow$

$E[X^2] \ge (E[X])^2$, which is exactly Jensen for $\Phi(x) = x^2$.

Problem 2. Let (X,Y) have joint density $f(x,y) = \begin{cases} 2 : & 0<y<x<1 \\ 0 : & \text{otherwise} \end{cases}$

(a) Find the marginal density of $X^{a.1}$ and compute $\mathbb{E}[X]^{a.2}$

(b) Find the conditional density $f_{Y|X}(y|x)$

(c) Compute $\text{Cov}(X,Y)$

(a) The density is nonzero on $R = \{(x,y) : 0<y<x<1\}$ Means: {x ranges from 0 to 1}, {for each fixed $x \in (0,1)$ y ranges from 0 up to x.

Before anything else, check it really integrates to 1 (so it's a valid joint PDF):

$$\int_0^1 \int_0^x 2\,dy\,dx = \int_0^1 \left(2y\Big|_0^x\right)dx = \int_0^1 2x\,dx = x^2\Big|_0^1 = 1$$ So it's valid (Wasserman Def 2.19)

Marginal density $f_X(x)$ and $E[X]$

(a.1) Compute $f_X(x)$ By definition of marginal density, we integrate the joint density over all y values that are allowed for that x. For a given $x \in (0,1)$, allowed y are $0<y<x$. So:

$$f_X(x) = \int_0^x f_{X,Y}(x,y)\,dy = \int_0^x 2\,dy = 2y\Big|_0^x = 2x, \quad 0<x<1$$ and $f_X(x) = 0$ otherwise.

So $f_X(x) = \begin{cases} 2x, & 0<x<1 \\ 0, & \text{otherwise} \end{cases} \longrightarrow f_X(x) = 2x$ on $(0,1)$

(a.2) Compute $E[X]$ use $E[X] = \int x f_X(x)\,dx \rightarrow E[X] = \int_0^1 x(2x)\,dx = \int_0^1 2x^2\,dx = 2\left(\frac{x^3}{3}\Big|_0^1\right) = 2 \cdot \frac{1}{3} = \frac{2}{3}$

$\rightarrow \mathbb{E}[X] = \frac{2}{3}$

(b) Using the conditional density formula: $f_{Y|X}(y|x) = \dfrac{f_{X,Y}(x,y)}{f_X(x)}$

For $0<x<1$ and $0<y<x$

$f_{X,Y}(x,y) = 2$

$f_X(x) = 2x$ from (a.1)

So: $f_{Y|X}(y|x) = \frac{2}{2x} = \frac{1}{x}$, $0<y<x$ and it is 0 outside that range.

So: $f_{Y|X}(y|x) = \begin{cases} \frac{1}{x}, & 0<y<x,\ 0<x<1 \\ 0, & \text{otherwise} \end{cases}$ $\longrightarrow$ Quick check, must integrate $\boxed{\text{to } 1}$ over y $\int_0^x \frac{1}{x}\,dy = \frac{1}{x}y\Big|_0^x = \frac{1}{x} \cdot x = 1$

Then $f_{Y|X}(y|x) = \frac{1}{x}$ for $0<x<y$

(c) Compute $\text{cov}(X,Y)$. We will use $\text{cov}(X,Y) = E[XY] - E[X]E[Y]$ (Wasserman Theorem 3.19)

We already have $E[X] = \frac{2}{3}$. Now we need $E[Y]$ (c.1) and $E[XY]$ (c.2)

(c.1) Find $E[Y]$. From part (b) $f_{Y|X}(y|x) = \frac{1}{x}$ on $(0,x)$. Then we need to compute:

$$E[Y|X=x] = \int_0^x y \cdot \frac{1}{x}\,dy = \frac{1}{x}\int_0^x y\,dy = \frac{1}{x}\left(\frac{y^2}{2}\Big|_0^x\right) = \frac{1}{x} \cdot \frac{x^2}{2} = \frac{x}{2}.$$ Now apply iterated expectation:

$E[Y] = E(E[Y|X])$ (Wasserman Theorem 3.24) $\mathbb{E}[\mathbb{E}(Y|X)] = \mathbb{E}(Y)$ and $\mathbb{E}[\mathbb{E}(X|Y)] = \mathbb{E}(X)$

So $\mathbb{E}[Y] = \mathbb{E}\left(\frac{X}{2}\right) = \frac{1}{2}\mathbb{E}[X] = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$

(C.2) Find $E[XY]$, Again use conditional expectation: $\mathbb{E}[XY] = E(E[XY|X])$

Given $X=x$, the value x is a constant, so: $\mathbb{E}[XY|X=x] = x\,\mathbb{E}[Y|X=x] = x \cdot \frac{x}{2} = \frac{x^2}{2}$

Thus $E[XY] = E\left(\frac{X^2}{2}\right) = \frac{1}{2}E[X^2]$. So we compute $E[X^2]$ from the marginal $f_X(x) = 2x$

$$E[X^2] = \int_0^1 x^2(2x)\,dx = \int_0^1 2x^3\,dx = 2\left(\frac{x^4}{4}\Big|_0^1\right) = 2\cdot\frac{1}{4} = \frac{1}{2} \longrightarrow E[XY] = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$$

(C.2)

Replacing in covariance formula

$$\text{cov}(X,Y) = E[XY] - E[X]E[Y] = \frac{1}{4} - \left(\frac{2}{3}\right)\left(\frac{1}{3}\right) = \frac{1}{4} - \frac{2}{9} \rightarrow \text{Cov}(X,Y) = \frac{9}{36} - \frac{8}{36} = \frac{1}{36}$$

Then $\text{Cov}(X,Y) = \frac{1}{36}$

Problem 3

Let $X \sim \text{Exponential}(\lambda)$ and define $Y = \log X$

(a) Find the PDF of Y

(b) Can the PDF of Y be written in the form $f_Y(y|\theta) = g(y\ \theta)$ for some function g and parameter theta. If so, show it. Note: If you can write the PDF that way then it belongs to a "location family"

(c) Find $\mathbb{E}[e^Y]$ [Not be confused with exponential family]

(a) First we need to identify the support of Y. Since $X \sim \text{Exponential}(\lambda)$, we know $X > 0$. So $Y = \text{Log}X$ can take any real value: $Y \in (-\infty, \infty)$.

Now lets check monotoniaty and find the inverse. let $g(x) = \text{Log}\,x$. on $x>0$, g is strictly increasing, so we may use the monotone transformation theorem (4_transformations.pdf page 4)

Now lets solve $y = \log x$ for x $\quad x = e^y \rightarrow g^{-1}(y) = e^y$. Then differentiate:

$\frac{d}{dy}g^{-1}(y) = \frac{d}{dy}(e^y) = e^y$. Now we need to apply the transformation theorem $f_Y(y) = f_X(e^y)|e^y|$

But $e^y > 0$, so $|e^y| = e^y$. Using the exponential PDF $f_X(x) = \lambda e^{-\lambda x}$ (5_common_Distributions.pdf Page 10)

$$f_Y(y) = \lambda e^{-\lambda(e^y)} \cdot e^y \longrightarrow f_Y(y) = \lambda e^y \exp(-\lambda e^y),\ -\infty < y < \infty$$

(b) What we already know from part (a). We are given $X \sim \text{Exponential}(\lambda)$ so $f_X(x) = \lambda e^{-\lambda x},\ x>0$ from _common Distributions Also $Y = \log X$. since $\log(\cdot)$ is strictly increasing on $(0,\infty)$, we can use the 1-1 transformation rule. $f_Y(y) = f_X(x(y))\left|\frac{dx}{dy}\right| \longrightarrow$ Here $x = e^y$ and $\frac{dx}{dy} = e^y$ so...

$$f_Y(y|\lambda) = f_X(e^y)\cdot e^y = \lambda e^{-\lambda e^y} e^y = \lambda e^y e^{-\lambda e^y},\ y \in \mathbb{R}$$

Now let's try to rewrite $f_Y(y|\lambda)$ as $g(y-\theta)$. Starting from the density $f_Y(y|\lambda) = \lambda e^y e^{-\lambda e^y}$

The trick is to absorb λ into a shift inside a log, because location forms look like "y minus something".

Notice: $\lambda e^y = e^{\log(\lambda)} e^y = e^{y+\log(\lambda)}$. Also, $\lambda e^y = e^{y+\log(\lambda)} \rightarrow e^{-\lambda e^y} = e^{-e^{y+\log(\lambda)}}$

So:

$$f_Y(y|\lambda) = e^{y+\log(\lambda)} e^{-e^{y+\log(\lambda)}}$$

Now define the location parameter $\theta = -\log(\lambda)$ Then $y-\theta = y-(-\log\lambda) = y+\log\lambda$

Let $u = y-\theta$. Then the density becomes $f_Y(y|\lambda) = e^u e^{-e^u}$

Now we need to identify $g(\cdot)$

We have shown: $f_Y(y|\lambda) = g(y-\theta)$ with $\theta = -\log(\lambda)$ and $\boxed{g(u) = e^u e^{-e^u},\ u \in \mathbb{R}}$

Lets check if g is a valid PDF. For that we need to verify $\int_{-\infty}^{\infty} g(u)du = 1 \rightarrow$

$\int_{-\infty}^{\infty} e^u e^{-e^u} du$. Let $t = e^u \rightarrow dt = e^u du$. when $u \rightarrow -\infty$, $t \rightarrow 0$. so:
when $u \rightarrow \infty$, $t \rightarrow \infty$

$$\int_{-\infty}^{\infty} e^u e^{-e^u} du = \int_0^{\infty} e^{-t} dt = 1$$ so g is indeed a density. So,

Yes. The PDF can be writen as $f_Y(y|\theta) = g(y-\theta)$ by taking $\theta = -\log(\lambda)$, $g(u) = e^u e^{-e^u}$ then

$$f_Y(y|\lambda) = \lambda e^y e^{-\lambda e^y} = g(y-\theta)$$

(c) Find $\mathbb{E}[e^Y]$ when $X \sim \text{Exponential}(\lambda)$ and $Y = \log X$

Lets Rewrite e^Y in terms of X.

Given $Y = \log X$ (with $X > 0$ for an exponential rv), we have the identity: $e^Y = e^{\log X} = X$

So the expectations becomes: $\mathbb{E}[e^Y] = \mathbb{E}[X]$

Lets compute $\mathbb{E}[X]$ for $X \sim \text{Exponential}(\lambda)$.

We have two methods

(A) From 5_Common_Distribution. (page 10) $\mathbb{E}[X] = \frac{1}{\lambda} \rightarrow \mathbb{E}[e^Y] = \frac{1}{\lambda}$

(B) Using the definition of expectation. Because X is continuos, $\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$

From Wasserman Def 3.1. For an exponential (λ) rv,

$$\mathbb{E}[X] = \int x \, dF(x) = \begin{cases} \sum_x x f(x) & \text{if } X \text{ is discrete} \\ \int x f(x) dx & \text{if } X \text{ is continuos} \end{cases} \quad ; \; f_X(x) = \lambda e^{-\lambda x}, \; x \geq 0$$

So $E[X] = \int_0^\infty x \lambda e^{-\lambda x} dx$ now compute the integral by integration by parts:

Let $u = x \rightarrow du = dx$

$dv = \lambda e^{-\lambda x} dx \rightarrow v = \int \lambda e^{-\lambda x} dx = -e^{-\lambda x}$

Then:

$$\int_0^\infty x \lambda e^{-\lambda x} dx = x(-e^{-\lambda x}) \Big|_0^\infty - \int_0^\infty (-e^{-\lambda x}) dx = -x e^{-\lambda x} \Big|_0^\infty + \int_0^\infty e^{-\lambda x} dx$$

Evaluate the boundary term:

At $x = 0$: $-x e^{-\lambda x} = 0$

As $x \rightarrow \infty$: $x e^{-\lambda x} \rightarrow 0$, so $-x e^{-\lambda x} \rightarrow 0$.

Thus the boundary term is $0 - 0 = 0$. $\longrightarrow E[X] = \int_0^\infty e^{-\lambda x} dx = -\frac{1}{\lambda} e^{-\lambda x} \Big|_0^\infty = 0 - \left(-\frac{1}{\lambda}\right) = \frac{1}{\lambda}$

Therefore $E[e^Y] = E[X] = \frac{1}{\lambda}$